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Unit 2: System Design and Troubleshooting

MCQs and Short Questions — Exam Practice

Section A: Multiple Choice Questions

Circle the correct answer. Each question has one correct answer.

Topic: Analog and Digital Signals / ADC / DAC

1. Which of the following is an example of an analog signal?

- a) Binary data stored in a computer
 - b) **Sound waves (voice signal)**
 - c) Data transmitted over a digital network
 - d) Pixels on a computer screen
-

2. Which of the following best describes a digital signal?

- a) A signal that changes smoothly and continuously
 - b) A signal with infinite possible values
 - c) **A signal that has only two values: 0 and 1**
 - d) A signal that can only be used in radios
-

3. What does ADC stand for?

- a) Analog Data Computing
 - b) **Analog to Digital Converter**
 - c) Automatic Digital Circuit
 - d) Advanced Data Conversion
-

4. What is the purpose of a DAC?

- a) To convert digital signals to binary

- b) To convert analog signals to digital signals
 - c) **To convert digital signals back to analog signals**
 - d) To store data in digital format
-

5. Why is ADC and DAC conversion important?

- a) It makes computers run faster
 - b) It reduces the cost of electronic devices
 - c) **It enables data processing, storage, and transmission with less noise**
 - d) It increases the number of bits in a signal
-

6. Which device uses DAC to produce output that humans can hear?

- a) Microphone
 - b) Keyboard
 - c) Scanner
 - d) **Speaker**
-

7. When a person speaks into a microphone, which conversion takes place?

- a) DAC — Digital to Analog
 - b) **ADC — Analog to Digital**
 - c) Binary to Hexadecimal
 - d) Decimal to Octal
-

8. Which of the following is a characteristic of an analog signal?

- a) Discrete values only
 - b) Only two possible values
 - c) **Continuous and smooth variation over time**
 - d) Represented using 0s and 1s
-

Topic: Boolean Algebra — AND, OR, NOT

9. Which of the following Boolean expressions represents the AND operation?

- a) $A + B$

- b) $\bar{A}$
 - c) $A \cdot B$
 - d) $A \oplus B$
-

10. Which of the following Boolean expressions represents the OR operation?

- a) $A \cdot B$
 - b) $\bar{A}$
 - c) $A + B$
 - d) $A \oplus B$
-

11. What is the output of the AND operation when $A = 1$ and $B = 0$?

- a) 1
 - b) 0
 - c) Undefined
 - d) 2
-

12. What is the output of the OR operation when $A = 0$ and $B = 0$?

- a) 1
 - b) Undefined
 - c) 2
 - d) 0
-

13. What is the output of the OR operation when $A = 1$ and $B = 1$?

- a) 0
 - b) 2
 - c) 1
 - d) Undefined
-

14. What is the result of the NOT operation when $A = 0$?

- a) 0
- b) 1

- c) 2
 - d) Undefined
-

15. How many inputs does the NOT gate accept?

- a) Two
 - b) Three
 - c) Zero
 - d) **One**
-

16. In Boolean algebra, what is the result of $1 + 1$ in the OR operation?

- a) 2
 - b) 0
 - c) 10
 - d) **1**
-

17. The AND operation gives an output of 1 only when:

- a) At least one input is 1
 - b) Both inputs are 0
 - c) **Both inputs are 1**
 - d) Neither input is 1
-

18. The OR operation gives an output of 0 only when:

- a) Both inputs are 1
 - b) At least one input is 1
 - c) One input is 1 and the other is 0
 - d) **Both inputs are 0**
-

19. Which symbol is used to represent the AND operation in Boolean algebra?

- a) +
- b) $\oplus$
- c) $-$

- d) $\cdot$ (**dot**)
-

20. Which symbol is used to represent the OR operation in Boolean algebra?

- a) $\cdot$
 - b) $\oplus$
 - c) $+$
 - d) $\times$
-

Topic: Logic Gates

21. Which logic gate outputs true only when both inputs are true?

- a) OR gate
 - b) XOR gate
 - c) NOT gate
 - d) **AND gate**
-

22. Which logic gate outputs true when at least one input is true?

- a) AND gate
 - b) NOT gate
 - c) **OR gate**
 - d) NAND gate
-

23. The NAND gate is a combination of which two gates?

- a) OR gate + NOT gate
 - b) XOR gate + NOT gate
 - c) **AND gate + NOT gate**
 - d) OR gate + AND gate
-

24. What does the NAND gate output when both inputs are 1?

- a) 1
- b) Undefined
- c) 2

- d) **0**
-

25. The XOR gate outputs true only when:

- a) Both inputs are true
 - b) Both inputs are false
 - c) **Exactly one input is true**
 - d) Neither input is true
-

26. What is the output of a XOR gate when $A = 1$ and $B = 1$?

- a) 1
 - b) Undefined
 - c) 2
 - d) **0**
-

27. What is the output of a XOR gate when $A = 1$ and $B = 0$?

- a) 0
 - b) **1**
 - c) 2
 - d) Undefined
-

28. Which gate is known as the universal gate because it can implement any Boolean function?

- a) OR gate
 - b) XOR gate
 - c) AND gate
 - d) **NAND gate**
-

29. A NOT gate is also called:

- a) Buffer
- b) XOR gate
- c) **Inverter**
- d) Multiplexer

30. Which logic gate would you use to implement the Boolean function $F = A \cdot B$?

- a) OR gate
 - b) NOT gate
 - c) XOR gate
 - d) **AND gate**
-

Topic: Boolean Functions and Truth Tables

31. What is a truth table used for?

- a) To store binary data
 - b) To convert decimal to binary
 - c) **To show the output of a Boolean function for all possible input combinations**
 - d) To draw logic diagrams
-

32. How many rows does a truth table for two binary variables (A and B) have?

- a) 2
 - b) 3
 - c) **4**
 - d) 8
-

33. How many rows does a truth table for three binary variables (A, B, C) have?

- a) 4
 - b) 6
 - c) 3
 - d) **8**
-

34. For the Boolean function $F(A, B) = A \cdot B$, what is the output when $A = 1$ and $B = 1$?

- a) 0
 - b) **1**
 - c) 2
 - d) Undefined
-

35. For the Boolean function $F(A, B) = A + B$, what is the output when $A = 0$ and $B = 1$?

- a) 0
 - b) **1**
 - c) 2
 - d) Undefined
-

36. Who invented Boolean algebra?

- a) Alan Turing
 - b) John von Neumann
 - c) Charles Babbage
 - d) **George Boole**
-

37. Where was George Boole born?

- a) London, England
 - b) Manchester, England
 - c) **Lincoln, England**
 - d) Birmingham, England
-

38. Boolean algebra uses how many values?

- a) Ten (0–9)
 - b) Sixteen (0–F)
 - c) **Two (True and False)**
 - d) Eight (0–7)
-

Topic: Boolean Simplification Laws

39. According to the Identity Law, what is the result of $A + 0$?

- a) 0
 - b) 1
 - c) **A**
 - d) $\bar{A}$
-

40. According to the Identity Law, what is the result of $A \cdot 1$?

- a) 0
 - b) 1
 - c) $\bar{A}$
 - d) **A**
-

41. According to the Null Law, what is the result of $A + 1$?

- a) A
 - b) 0
 - c) $\bar{A}$
 - d) **1**
-

42. According to the Null Law, what is the result of $A \cdot 0$?

- a) A
 - b) 1
 - c) $\bar{A}$
 - d) **0**
-

43. According to the Complement Law, what is the result of $A + \bar{A}$?

- a) 0
 - b) A
 - c) $\bar{A}$
 - d) **1**
-

44. According to the Complement Law, what is the result of $A \cdot \bar{A}$?

- a) 1
 - b) A
 - c) $\bar{A}$
 - d) **0**
-

45. According to the Double Negation Law, what is the result of NOT(NOT A)?

- a) 0
 - b) 1
 - c) $\bar{A}$
 - d) **A**
-

46. De Morgan's first theorem states that $A + B$ is equal to:

- a) $A + B$
 - b) $\bar{A} \cdot \bar{B}$
 - c) $A \cdot B$
 - d) $\bar{A} + \bar{B}$
-

47. De Morgan's second theorem states that $A \cdot B$ is equal to:

- a) $\bar{A} \cdot \bar{B}$
 - b) $A + B$
 - c) $\bar{A} + \bar{B}$
 - d) $A \cdot B$
-

48. The Idempotent Law states that $A + A$ equals:

- a) 0
 - b) 1
 - c) $2A$
 - d) **A**
-

49. The Absorption Law states that $A + (A \cdot B)$ equals:

- a) B
 - b) $A \cdot B$
 - c) $A + B$
 - d) **A**
-

50. The Distributive Law states that $A \cdot (B + C)$ equals:

- a) $A \cdot B + C$
 - b) **$(A \cdot B) + (A \cdot C)$**
 - c) $A + B \cdot C$
 - d) $A \cdot B \cdot C$
-

Topic: Troubleshooting

51. What is the first step in the systematic process of troubleshooting?

- a) Establish a Theory of Probable Cause
 - b) Implement the Solution
 - c) **Identify the Problem**
 - d) Document Findings, Actions, and Outcomes
-

52. Which step involves coming up with a theory about what might be causing a problem?

- a) Test the Theory to Determine the Cause
 - b) **Establish a Theory of Probable Cause**
 - c) Implement the Solution
 - d) Verify Full System Functionality
-

53. After implementing a solution, what is the next step in the troubleshooting process?

- a) Document Findings, Actions, and Outcomes
 - b) Test the Theory to Determine the Cause
 - c) **Verify Full System Functionality**
 - d) Establish a Plan of Action to Resolve the Problem
-

54. What is the last step in the systematic troubleshooting process?

- a) Verify Full System Functionality
 - b) Implement the Solution
 - c) Establish a Theory of Probable Cause
 - d) **Document Findings, Actions, and Outcomes**
-

55. How many steps are in the systematic troubleshooting process?

- a) 5
 - b) 6
 - c) **7**
 - d) 8
-

56. Why is effective troubleshooting important for maintaining systems?

- a) It helps save money on repairs
 - b) It prevents the need for professional help
 - c) **It ensures systems operate smoothly and efficiently**
 - d) It allows for more frequent system updates
-

57. What does "downtime" mean in the context of computing?

- a) When a computer runs slowly
 - b) When a computer needs an update
 - c) When a computer screen dims
 - d) **When a computer system is not operational**
-

58. Which of the following is a common cause of computer overheating?

- a) Too many software applications installed
 - b) Weak internet connection
 - c) **Poor ventilation around the computer**
 - d) Low screen brightness
-

59. Which tool is used to diagnose RAM failures?

- a) CrystalDiskInfo
 - b) SMART
 - c) Disk Cleanup
 - d) **MemTest86**
-

60. Which tool is used to monitor hard drive health?

- a) MemTest86
 - b) Windows Memory Diagnostic
 - c) **CrystalDiskInfo**
 - d) Task Manager
-

61. A Blue Screen of Death (BSOD) is commonly associated with which hardware issue?

- a) Overheating processor
 - b) Loose cable connections
 - c) **Faulty RAM**
 - d) Disconnected keyboard
-

62. What should you do before replacing a hard drive?

- a) Format the new drive first
 - b) Reinstall the operating system
 - c) **Back up your data**
 - d) Update device drivers
-

63. Which of the following best describes "data integrity"?

- a) Keeping data stored in binary format
 - b) Encrypting all data files
 - c) **Ensuring data is accurate and reliable**
 - d) Compressing data to save space
-

64. The WannaCry ransomware attack in 2017 exploited which vulnerability?

- a) Weak passwords on user accounts
 - b) Outdated antivirus software
 - c) **Unpatched older Windows systems**
 - d) Slow internet connections
-

65. What is the quickest first fix to try when a computer is not responding?

- a) Replace the RAM
 - b) Reinstall the operating system
 - c) Format the hard drive
 - d) **Restart the computer**
-

66. Which of the following is an example of system maintenance software?

- a) Microsoft Word
 - b) Google Chrome
 - c) **Antivirus software**
 - d) VLC Media Player
-

67. What is the purpose of installing software patches?

- a) To add new features to games
 - b) To increase RAM capacity
 - c) **To fix bugs and protect against security vulnerabilities**
 - d) To speed up internet connection
-

68. Which of the following is a symptom of a failing hard drive?

- a) Screen flickering
 - b) Keyboard not responding
 - c) Mouse cursor moving erratically
 - d) **Strange clicking noises from the drive**
-

69. What does SMART stand for in hard drive diagnostics?

- a) System Monitoring and Recovery Tool
 - b) **Self-Monitoring, Analysis, and Reporting Technology**
 - c) Standard Memory and RAM Testing
 - d) Software Maintenance and Repair Toolkit
-

70. Troubleshooting helps improve computer security by:

- a) Installing more RAM
 - b) Replacing old keyboards
 - c) **Identifying weaknesses and security breaches**
 - d) Increasing screen resolution
-

Section B: Short Questions

Answer each question in 2–4 sentences in your own words.

Topic: Analog and Digital Signals / ADC / DAC

1. Define an analog signal. Give two examples from daily life.
 2. Define a digital signal. How is it different from an analog signal?
 3. What is ADC? Why is it needed in computing?
 4. What is DAC? Give one real-world example of where DAC is used.
 5. Why are digital signals preferred over analog signals for transmitting information over long distances?
 6. Describe what happens when a person speaks into a microphone and the sound reaches another person through a speaker. Which conversions take place?
-

Topic: Boolean Algebra — AND, OR, NOT

1. Define a Boolean function.
2. What are binary variables? What values can they take?
3. Explain the AND operation in Boolean algebra. When does it output 1?
4. Explain the OR operation in Boolean algebra. When does it output 0?
5. Explain the NOT operation. How many inputs does it take?
6. Draw the truth table for the AND operation with two inputs A and B.
7. Draw the truth table for the OR operation with two inputs A and B.
8. Draw the truth table for the NOT operation with one input A.
9. Write the Boolean expression for the AND operation between variables A and B.
10. Write the Boolean expression for the OR operation between variables A and B.

11. What is the significance of a truth table in digital logic?
 12. Construct the truth table for $F(A, B, C) = A \cdot B + \bar{A} \cdot C$. Show all intermediate columns.
-

Topic: Logic Gates

1. What is a logic gate? What is its relationship to Boolean operations?
 2. Describe the function of an AND gate. What must be true for its output to be 1?
 3. Describe the function of an OR gate with its truth table.
 4. Describe the function of a NOT gate with its truth table.
 5. What is a NAND gate? How is it formed? Write its truth table.
 6. What is an XOR gate? How does it differ from the OR gate?
 7. Write the truth table for the XOR gate.
 8. Give a real-life example that illustrates the working of an AND gate.
 9. Give a real-life example that illustrates the working of a NAND gate.
 10. Give a real-life example that illustrates the working of an XOR gate.
-

Topic: Boolean Functions and Logic Diagrams

1. What is a Boolean function? What types of values can its inputs and outputs take?
 2. Construct the truth table for $F(A, B) = A \cdot B$ and explain each row.
 3. Construct the truth table for $F(A, B) = A + B$ and explain each row.
 4. How is a Boolean function used in the Arithmetic Logic Unit (ALU) of a CPU?
 5. List three uses of Boolean functions in computers.
 6. What is a logic diagram? What does it represent?
 7. What are the steps to create a logic diagram from a Boolean function?
 8. Who invented Boolean algebra and in what year was he born?
-

Topic: Boolean Simplification

1. State the Identity Laws of Boolean algebra.
2. State the Null Laws of Boolean algebra.
3. State the Idempotent Laws of Boolean algebra.

4. State the Complement Laws of Boolean algebra.
 5. State De Morgan's Theorems and explain what they mean.
 6. State the Absorption Laws of Boolean algebra.
 7. State the Double Negation Law and give an example.
 8. State the Distributive Laws of Boolean algebra.
 9. Why is simplification of Boolean functions important in digital circuit design?
 10. Simplify the Boolean expression: $F = A \cdot A$.
 11. Simplify the Boolean expression: $F = A + 1$.
 12. Simplify the Boolean expression: $F = A + (A \cdot B)$ using the Absorption Law.
-

Topic: System Troubleshooting

1. What is troubleshooting? Why is it important?
 2. List the seven steps of the systematic troubleshooting process in order.
 3. What is the first step in the systematic process of troubleshooting?
 4. After identifying a problem, what is the next step in troubleshooting?
 5. Describe the importance of testing a theory during the troubleshooting process.
 6. Explain what the "Implement the Solution" step means in troubleshooting.
 7. Why is it necessary to verify full system functionality after implementing a solution?
 8. Why is it important to document findings, actions, and outcomes after troubleshooting?
 9. What is downtime? How does effective troubleshooting help prevent it?
 10. What is data integrity? How can troubleshooting help protect it?
 11. How does troubleshooting help improve computer security?
 12. How can regular troubleshooting extend the life of computer equipment?
-

Topic: Hardware Issues and Maintenance

1. What are three common hardware issues that can affect a computer system?
2. What is the recommended solution for preventing computer overheating?
3. List two common symptoms of RAM failure.
4. List two common symptoms of hard drive failure.

5. What is MemTest86 used for?
 6. What is CrystalDiskInfo used for?
 7. What does SMART stand for? What is it used for?
 8. What steps should you follow when replacing a RAM module?
 9. What steps should you follow when replacing a hard drive?
 10. Why is it important to back up data before replacing a hard drive?
-

Topic: Software Maintenance and Security

1. Why is it important to keep software up to date?
 2. What is a software patch? Why are patches important?
 3. What is a software conflict? How can it be resolved?
 4. How can you create a strong password? What should it include?
 5. What was the WannaCry ransomware attack? What lesson does it teach about software updates?
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End of Unit 2 MCQs and Short Questions